

(a) $t = 0.00 \text{ s}, 0^\circ$ (b) $t = 2.38 \text{ s}, 3^\circ$ (c) $t = 4.21 \text{ s}, 12^\circ$ (d) $t = 4.38 \text{ s}, 61^\circ$ (e) $t = 4.42 \text{ s}, 98^\circ$

